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Department of Computer Science and Engineering

Societal Project - Review 0

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EduSolve

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Abstract

EduSolve connects students with real-world community problems and opportunities. It matches problems with students based on their skills and interests. The platform supports communication, approval, and project collaboration. It helps students gain practical and meaningful project experience. It also enables communities and organizations to benefit from student-led solutions. **This creates a structured bridge between academic learning and societal problem-solving.**

Report problem Societal Problem

Lack of a structured platform connecting students with real-world community problems.

Groups Stakeholders

Students, communities, NGOs, local organizations, and problem owners.

Computer Technology Approach

A web-based platform with skill matching, recommendations, and collaborative project workspaces.

Trending up Expected Impact

Meaningful learning opportunities for students and potential solutions for real-world community problems.



Introduction



School

Educational Mandate (NEP 2020)

Education is increasingly expected to develop practical skills, critical thinking, and experiential learning. India's NEP 2020 strongly emphasizes inquiry-based and problem-solving-oriented learning.

Help outline

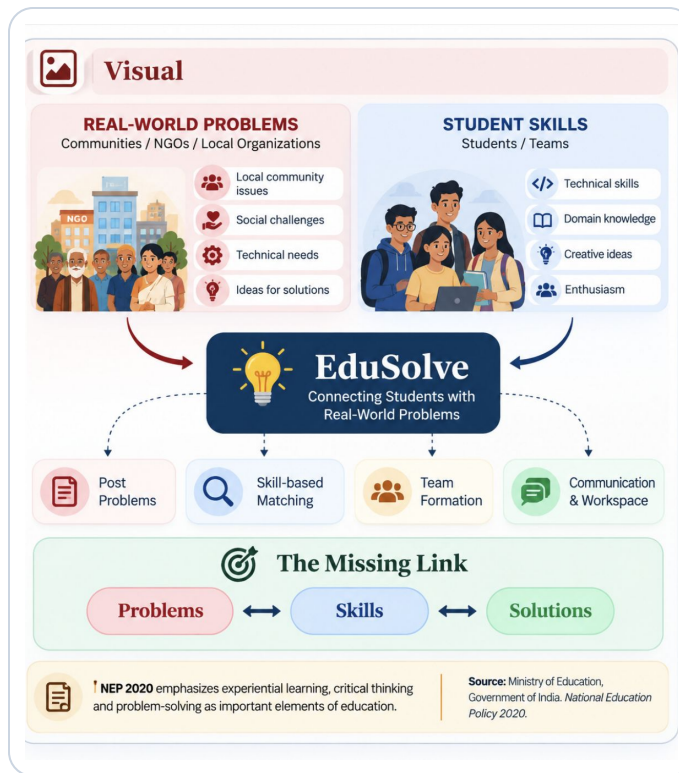
The Core Challenge

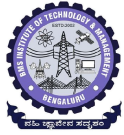
Students lack structured avenues to discover real-world problems for their technical skills. Concurrently, NGOs, local organizations, and communities face practical issues needing technical solutions.

Hub

The EduSolve Solution

EduSolve proposes a digital platform bridging both sides—helping problem owners post challenges while empowering students to discover tailored opportunities based on skills and interests.





Motivation

Find in page Field Observation

Students:

Many students have technical skills but find it difficult to identify suitable real-world problems for project work.

Problem owners:

Community groups or local organizations may have practical problems but may not know how to connect with students who have the required skills.

Article Data / Reports

Community Engagement

Higher education encourages students to work with communities on real-world issues.

Learning + Community

Students can use their knowledge and skills to help solve community problems.

Practical Exposure

Students are encouraged to gain practical experience through live projects and community/NGO collaboration.

Verified Why Us?

CSE Technical Skillset

Our team consists of Computer Science and Engineering students with skills in software development, web technologies, databases, and emerging AI-based technologies.

Practical & Social Impact

EduSolve allows us to apply these skills to a socially relevant problem while gaining practical experience through stakeholder interaction, prototype development, and validation.



Literature Survey

Sl No	Author(s) & Year	Title & Source	Approach / Method	Key Findings	Limitations / Gaps
[1]	Guo et al., 2020	A Review of Project-Based Learning in Higher Education — IJER	Review of 76 studies	PBL supports cognitive, affective and behavioral outcomes	Learning processes and final products need better evaluation
[2]	Natarajarathinam et al., 2021	Community Engagement in Engineering Education — JEE	Systematic review of 120 articles	Identified benefits and challenges of community engagement	Community-partner perspective was less studied
[3]	Celio et al., 2011	A Meta-Analysis of the Impact of Service-Learning on Students — JEE	Meta-analysis of 62 studies / 11,837 students	Gains in academic, social and civic outcomes	Outcomes depend on implementation practices
[4]	Urdaneta-Ponte et al., 2021	Recommendation Systems for Education — Electronics	Systematic literature review	Reviews recommendation techniques used in education	Focus is mainly on educational-resource recommendations
[5]	Knez et al., 2017	Implementation of Group Formation Algorithms in ELARS — iJET	Automatic student-group formation	Groups formed using student characteristics and preferences	Focuses on forming student groups, not matching students to external community problems



Research Gaps & Relevance

Gaps Identified

1. Limited connection to community problems

Many learning approaches focus on academic or institution-defined projects rather than creating a structured link with community problems.

2. Lack of skill-based problem matching

Students may have different skills and interests, but existing approaches lack a dedicated mechanism to match them with suitable real-world problems.

3. Limited end-to-end collaboration

There is a need for a structured process covering problem posting, student interest, selection, communication and collaborative project work.

Relevance to Present Context

Students

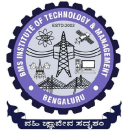
Apply technical knowledge to meaningful problems.

Communities

Bring their practical problems to a platform where students can contribute.

Institutions

Encourage experiential, project-based and community-connected learning.



Problem Statement



Students often possess technical and academic skills but lack a structured way to discover and collaborate on meaningful real-world problems, while communities and organizations may have practical problems that need student participation and technical support.

There is a need for a digital platform that can connect these problems with suitable students based on their skills and interests and support the collaboration process.

Stakeholders

- Students
- Communities / NGOs
- Organizations / Problem Owners
- Educational Institutions

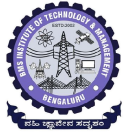
Evidence

Evidence to be collected through:

- Student & community surveys
- Field observations
- Relevant UGC / government reports

Scope

- Students & problem owners
- Problem posting & skill matching
- Student interest & approval
- Communication & workspace



Objectives

01

Survey & Need Assessment

50 / 10

Students & Problem Owners

1. To survey **50 students and 10 problem owners in Bengaluru** to assess the need for student–community project collaboration.

02

Prototype Development

EduSolve Platform

Interactive Digital Prototype

2. To develop an **EduSolve prototype** for problem posting, skill-based matching, student interest, approval and project collaboration.

03

Validation & Usability

$\geq 80\%$ Usability

Positive Feedback Target

3. To validate the prototype with **20 students and 5 problem owners** and achieve at least **80% positive usability feedback**.



SDG (Sustainable Development Goal)

SDG 4 • TARGET 4.4

Quality Education



Connection to the Project

Practical Skill Application

EduSolve helps students apply their technical and academic skills to real-world problems through practical projects and collaboration.

Target 4.4 Alignment

This directly supports skill development and practical learning, fully aligning the project with **SDG 4 Target 4.4** (Relevant skills for employment, decent jobs, and entrepreneurship).



Proposed Methodology & Work Plan



Week(s)	Planned Activity	Deliverable
1 – 4	Student survey, Research, Problem solving, Review I	Synopsis, slides
5 – 7	Data collection, System design, System workflow	Survey data, design
8 – 10	Implementation, Project building and testing	Prototype, results
11 – 12	Validation, Report, Review II	Report, paper draft



Expected Outcome

Space dashboard

Prototype / Model / Audit

Working EduSolve web prototype with problem posting, skill-based matching, student interest, owner approval, communication and project workspace.

Groups

Societal Impact

Connect students with real-world problems and provide practical learning opportunities through community collaboration.

Menu book

Publication

Research paper targeting a Scopus/Web of Science indexed conference or journal.

Verified

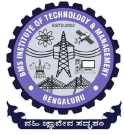
IPR & Report

Final project report and copyright/IPR submission, if applicable.



References

- [1] **P. Guo, N. Saab, L. S. Post, and W. Admiraal**, “A review of project-based learning in higher education: Student outcomes and measures,” *International Journal of Educational Research*, vol. 102, Art. no. 101586, 2020, doi: 10.1016/j.ijer.2020.101586.
- [2] **M. Natarajarathinam, S. Qiu, and W. Lu**, “Community engagement in engineering education: A systematic literature review,” *Journal of Engineering Education*, vol. 110, no. 4, pp. 1049–1077, 2021, doi: 10.1002/jee.20424.
- [3] **C. I. Celio, J. Durlak, and A. Dymnicki**, “A meta-analysis of the impact of service-learning on students,” *Journal of Experiential Education*, vol. 34, no. 2, pp. 164–181, 2011, doi: 10.1177/105382591103400205.
- [4] **M. C. Urdaneta-Ponte, A. Mendez-Zorrilla, and I. Oleagordia-Ruiz**, “Recommendation systems for education: Systematic review,” *Electronics*, vol. 10, no. 14, Art. no. 1611, 2021, doi: 10.3390/electronics10141611.
- [5] **T. Knez, M. Holenko Dlab, and N. Hoic-Bozic**, “Implementation of group formation algorithms in the ELARS recommender system,” *International Journal of Emerging Technologies in Learning*, vol. 12, no. 11, pp. 198–207, 2017, doi: 10.3991/ijet.v12i11.7238.



THANK YOU

Questions & Suggestions